



DEALER BEST PRACTICE SERIES

Haul Road Optimization

Maintenance and Repair Application Component Renewal (CRC) Component Life Management MARC Management

Haul Road Optimization0

1.0 Introduction1

2.0 Best Practice Description1

3.0 Implementation Steps5

4.0 Benefits5

5.0 Resources Required5

6.0 Supporting Attachments / References5

7.0 Related Best Practices5

8.0 Acknowledgements5

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1.0 Introduction

Properly designed and optimized haul roads can help to reduce cycle times, reduce fuel burn, reduce component wear, and increase profit. The haul road optimization process is designed to identify haul roads that don't meet optimum performance criteria and improve them to realize cycle time and fuel gains. The Global Mining CI team has developed a process to provide several optimized options for haul roads and provide estimates in the gains or benefits that could be realized with each option.

2.0 Best Practice Description

2.1 Opportunity Recognition

The best opportunities for haul road optimization lie in permanent haul roads with up hill hauls. The optimization process takes some time, and haul road work may not be able to be started right away, so selected roads need to be part of the mine's long term plans.

Optimization candidates can be identified by:

- o Containing slopes with greater than 10% grade
- o Segments of non-constant grades
- o Segments where operators are unable to maintain full throttle.
- o Segments in which transmission downshifts to 1st gear
- o Poorly designed intersections or switchbacks



Figure 1: Example of mine with series of uphill roads, intersections and switchbacks

2.2 Data Collection

The data collection can be done by onsite personal with or without Caterpillar Dealer or Global Mining personal present. Once a target haul road has been identified two types of data are required for the optimization process; GPS data and truck Datalogger data. Mine production and cost data is also required for the financial models.

2.2.1 GPS Data – The GPS data should be taken on the target road with the hardware used on site. This insures the data can be merged with the site property maps without data correction. Data should be from production run only and should not go through scales or any other short term deviation

Data points should be taken in 5 lanes – one on the outside edge of each side, one down the middle (Figure 1, red lines), and one in the middle of each lane (Figure 1, blue lines). The

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Haul Road Optimization	DATE	CHG NO	NUMBER
	2/16/2015	02	0409-2.02-1165

middle of the loaded haul lane will be the data used for optimization, the others will be used to define the road in the mine's programs to find fill and cut volumes.

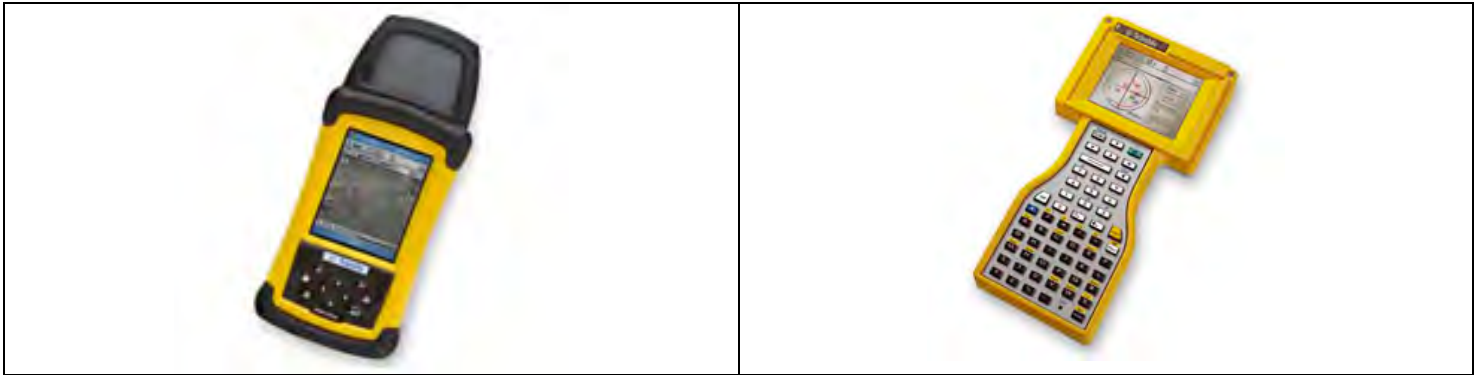


Figure 2: Example of handheld high precision GPS units

Data points should be taken every 3-5 feet (1 to 2 meters). This will provide coverage of all road surfaces that the truck may feel, and allow for better optimization in road transition areas. If data points are more than 10-20 feet (4 to 6 meters) the optimization program cannot efficiently optimize the road surface.

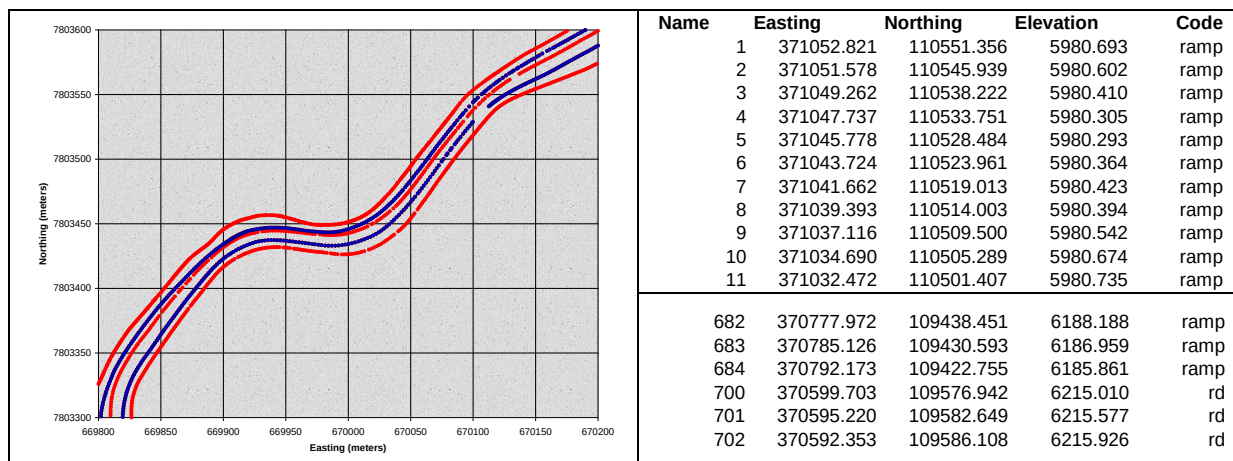


Figure 3: Example of DGPS data

It is advisable that the surveyor adds comments to the GPS waypoints about each lane, as shown in Figure 3 (right)

2.2.2 Datalogger Data –

Datalogger data should be taken at least 5 times from at least 3 different trucks. All three trucks should have good struts and the oil and grease should be recently recharged. The analysis of all these Dataloggers will help eliminate variations in driver performance from the study. The data should be watched for anomalies caused by haul road traffic, or serious driver deviation

Data should be taken from normal production runs, not runs where the truck stops for scales. Going through the scales adds distance to the run, and affects the speed profile as the truck slows to enter the scales and accelerates out. If the drivers are aware a study is taking

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Haul Road Optimization	DATE 2/16/2015	CHG NO 02	NUMBER 0409-2.02-1165
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place, they should know the study is of the haul road and not how they're driving, so they should drive as they always do.

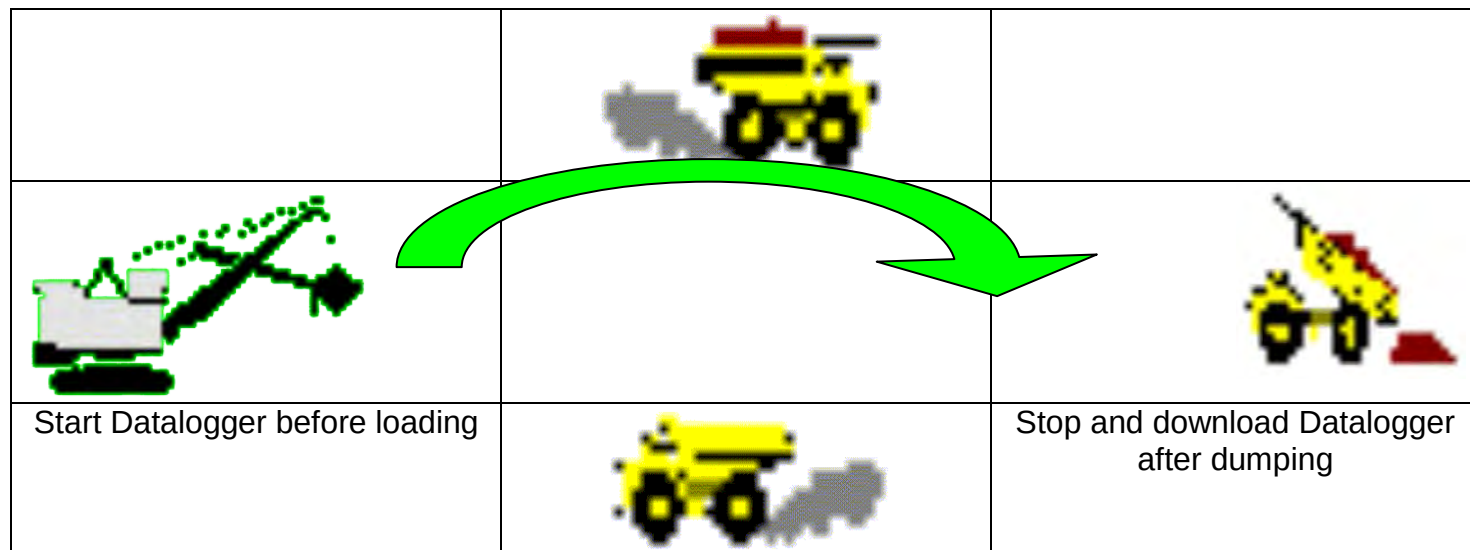


Figure 4: Datalogger setup and download

The Datalogger data should be downloaded along with the VIMS payload and event data.

2.2.3 Production and Cost Data –

Collect production and cost numbers from the mine in order to build the financial model, this includes:

- o Target Payload (ton)
- o Scheduled Hours: (hrs/day)
- o On-site fuel cost: (US\$/gal)
- o Operator efficiency (%)
- o Utilization? (%)
- o Striping ratio (tons:tons or yd3:ton)
- o Ore Grade (oz/ton or lb/ton)
- o Incremental margin on commodity price (US\$/lb or oz)
- o Ore Recovery Factor (%)

2.3 Road Optimization

The data gathered from the site is then sent to the Global Mining CI group. The CI group will optimize the GPS data and provide several options with varying levels of optimization. The most optimized roads will provide the most benefit, but will also require the most amount of road work. Estimated cycle times and financial benefits will also be provided for each design.

The CI group may also provide other recommended changes to the haul road, including possible redesigns that would reduce cycle times, super-elevations required to maintain speed around corners and cross fall recommendations to ensure proper drainage.

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Haul Road Optimization	DATE 2/16/2015	CHG NO 02	NUMBER 0409-2.02-1165

2.4 Presentation

The optimized GPS data can be loaded into either the mine's mapping software or into a 3D program to produce a 3D representation of the road. Mine software can also get estimates of the material that needs to be moved for the road work required.

A presentation should be prepared and presented to the mine management that will be making road change decisions. Road options, cycle time changes and financial benefits should be provided for consideration.

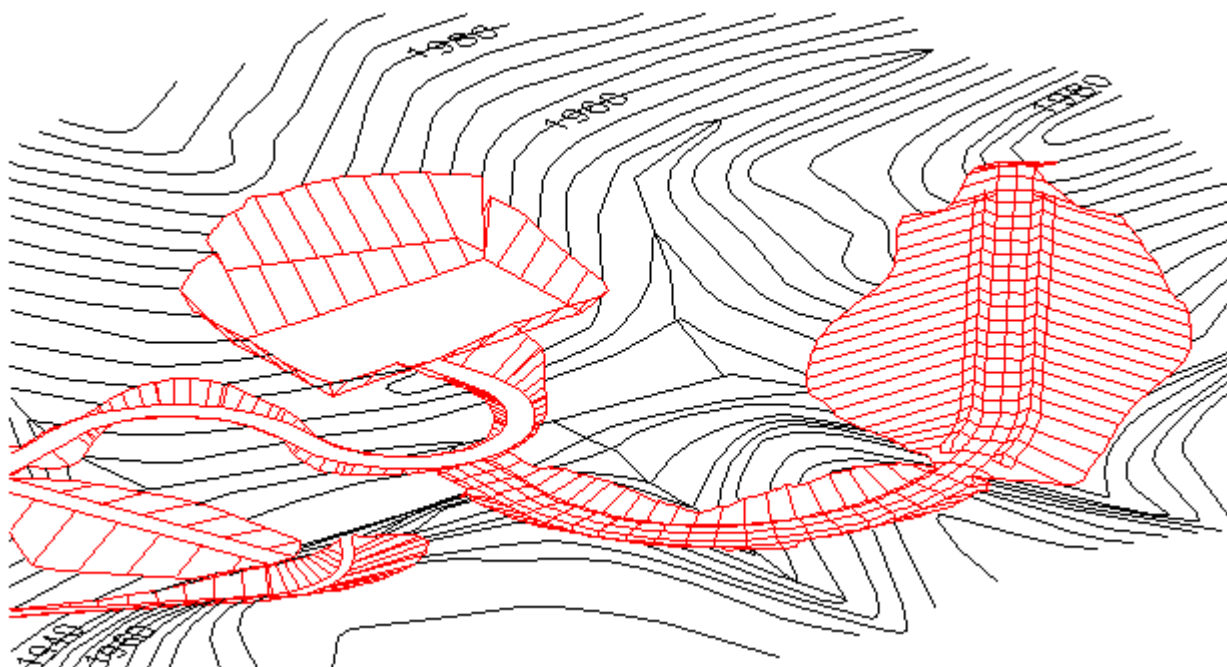


Figure 5: Example of road design using a 3D mapping software

2.5 Follow Up

Once an optimized option has been selected, mine personnel should be worked with to come up with a time table for road work to begin. If the site does not currently employ any systems such as CAES, the options can be presented to them as a way to improve their road work efficiency, speed and accuracy.

When road work is completed new cycle time and GPS data should be collected so the actual realized gains can be measured and reported. The GPS data can be sent back to the CI team and any gains from further optimization can be looked at.

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Haul Road Optimization	DATE 2/16/2015	CHG NO 02	NUMBER 0409-2.02-1165
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3.0 Implementation Steps

1. Haul roads that would benefit from optimization should be identified. The possibility of improvements should be discussed with the mine management.
2. Data should be collected and sent to the Cat Global Mining CI group for optimization.
3. When the haul road data has been optimized and returned a presentation should be prepared and presented to the mine management.
4. When an optimized option has been chosen, a road work plan and timeline should be discussed with the mine. Options for additional systems that would aid in the road work should also be discussed.
5. Once road work is complete new data should be collected and analyzed for actual gains.

4.0 Benefits

Optimized haul roads can provide substantial gains in cycle times, depending on the state, length and grade of the current haul road. Reduction in cycle times can lead to either an increase in production or a reduction in O&O costs.

An optimization process at one mine resulted in a 9% improvement in cycle time on a 3 mile haul road. Additional gains can be seen in component life gains from reduced hard shifting along the haul road.

5.0 Resources Required

- Mine site Surveyor – Current road to be surveyed using mine equipment
- CI team – Provide optimized options and estimated gains.
- Dealer support – Collect data to be sent to CI team, present optimized options to the mine management, and to follow up on road work to be completed.

6.0 Supporting Attachments / References

7.0 Related Best Practices

1106-1.1-1035	Cost Reduction Through Haul Road Management Process
1107-1.2-1094	Road Defect Management Training Exercise

8.0 Acknowledgements

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